

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457227.9055 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.061
Transit depth (Rp/Rs)^2	2.23 +/- 1.83 [%]
Orbital Inclination (inc)	84.31 +/- 1.91
Airmass coefficient 1 (a1)	0.98 +/- 0.61
Airmass coefficient 2 (a2)	0.06 +/- 1.41
Scatter in the residuals of the lightcurve fit is	63.26 %
Best Comparison Star	None
Optimal Aperture	0.87
Optimal Annulus	6.75
Transit Duration (day)	0.03 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.149 ± 0.061 vs 0.1592 (deviation 0.17σ)
Duration fit vs published	0.03 d vs 0.0946 d (deviation 1.96σ)
Mid-time O-C	-6.1 min
Depth SNR	2.4
Residual scatter	63.26 %

Light curve

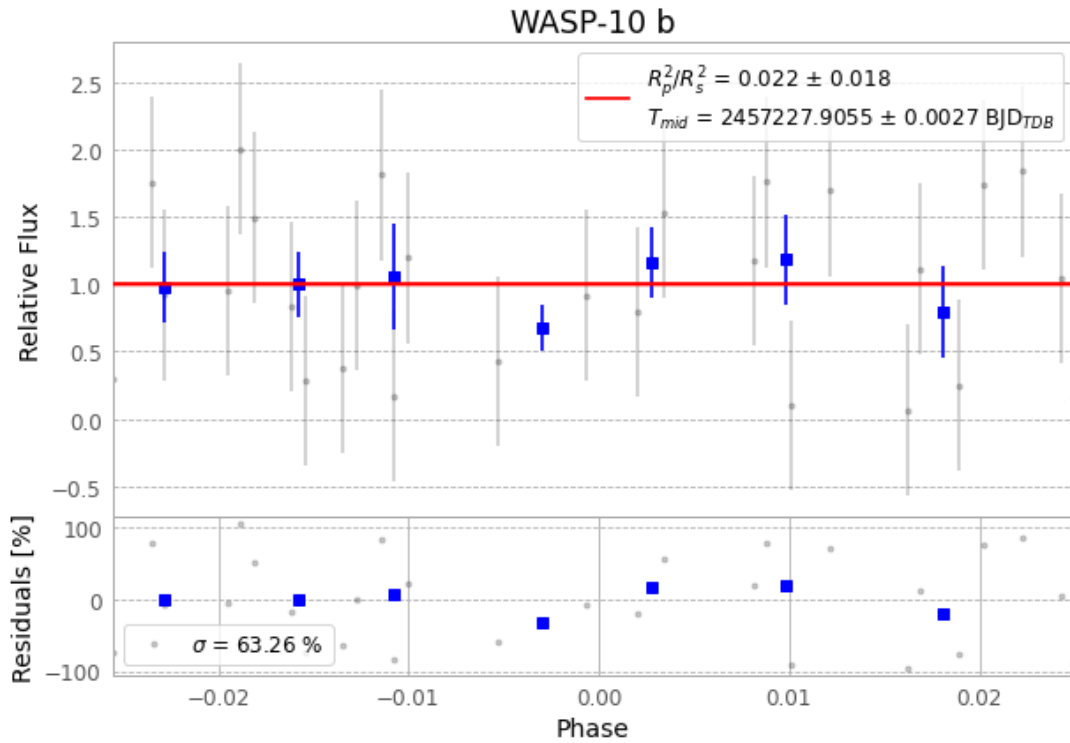


Figure 1 — FinalLightCurve_WASP-10 b_2015-07-24.png (SHA-256 6049d8d062b2b04a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [248, 360], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c8335dcf1cc703ab...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 341f7e9

Data provenance

77 FITS frames (51 MB), WASP-10150724074512.FITS → WASP-10150724113312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-150724.log (238840dfd1c5...), FinalLightCurve_WASP-10 b_2015-07-24.png (6049d8d062b2...), FinalLightCurve_WASP-10 b_2015-07-24.csv (b82c4636f6f9...)

Compute resources

Wall time 151.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 13:51Z.